The Making of a Scientist

The big idea

The Making of a Scientist by **Robert W. Peterson** describes how **Richard H. Ebright**, a boy fascinated by butterflies, grew into a famous scientist. It shows that **curiosity, hard work and encouragement** make a scientist.

Summary

- **Childhood:** Ebright grew up in **Reading, Pennsylvania (USA)**. He was an only child; his father died when he was in third grade, and his **mother** became his biggest support — taking him on trips, buying him telescopes, microscopes and cameras.
- **Collecting:** From kindergarten he collected **butterflies**, rocks, fossils and coins. By the second grade he had collected **all 25 species** of butterflies found around his hometown.
- **The turning point:** His mother gave him a book called **"The Travels of Monarch X"**, about the migration of monarch butterflies to Central America. It invited readers to help study migration by **tagging** butterflies for **Dr Frederick A. Urquhart** of the **University of Toronto**, Canada. Ebright raised and tagged thousands of monarchs.
- **Losing a science fair:** In seventh grade he entered a county science fair with slides of frog tissue but **lost**. He realised winners had done **real experiments**, not just displays.
- **Science fair projects:**
  - **Eighth grade:** he tested a theory that **viceroy butterflies copy monarchs** because birds avoid monarchs (they taste bad). The project did not win but led to further work.

- **Ninth/tenth grade:** he discovered the purpose of the **twelve tiny gold spots** on a monarch **pupa** — they produce a **hormone** necessary for the butterfly's full development. This won him prizes and recognition at science fairs.

- **Later research:** In his senior year he grew cells from a monarch's wing in culture and showed they would divide and become normal butterfly scales only if fed the hormone from the gold spots. These studies led him to a **new theory about how cells read their DNA blueprint** — the instructions inside cells. At **Harvard** he continued his research and became a respected scientist.
- **Other interests:** He was also a champion debater and public speaker, a canoeist and an expert photographer.

THREE QUALITIES OF A SCIENTIST (FROM THE TEXT)

1. **A first-rate mind.**
2. **Curiosity.**
3. **The will to win for the right reasons** — he did everything he could do well, not just to win.

EBRIGHT DID NOT WIN HIS FIRST SCIENCE FAIR

He **lost** in seventh grade, and that failure taught him to do real experiments.

Themes and value points

- Curiosity and observation are the roots of science.
- Encouragement from parents and teachers matters.
- Failure can be a teacher; persistence pays.
- Hobbies can lead to careers.

LIKELY QUESTIONS

How did Ebright's mother help him? How did "The Travels of Monarch X" change his life? What lesson did he learn from losing the science fair? What qualities make a scientist? What was his theory about cells?

Quick recap

- Richard H. Ebright; author Robert W. Peterson.
- Mother's encouragement; 25 butterfly species; "The Travels of Monarch X"; Dr Urquhart.
- Lost 7th-grade fair → real experiments: viceroy mimicry; gold spots and hormone.
- Theory of how cells read DNA; first-rate mind, curiosity, will to win for the right reasons.

Practice questions

1. Which book opened the world of science to Ebright?

   (a) On the Origin of Species (b) The Travels of Monarch X (c) A Brief History of Time (d) The Butterfly Book

2. Who was Dr Frederick A. Urquhart?

   (a) Ebright's teacher (b) His father (c) A scientist at the University of Toronto studying monarch migration (d) A Harvard professor

3. What lesson did Ebright learn from losing the seventh-grade science fair?

   (a) Science is boring (b) Butterflies are not important (c) He should stop entering fairs (d) He needed to do real experiments, not just displays

4. What did Ebright discover about the gold spots on a monarch pupa?

   (a) They are just decoration (b) They produce a hormone needed for full development (c) They scare birds away (d) They change colour with weather

5. Ebright's eighth-grade project tested the idea that

   (a) viceroy butterflies copy monarchs (b) monarchs live only a week (c) birds love monarchs (d) caterpillars cannot see

6. Who was Ebright's greatest supporter as a child?

   (a) His father (b) Dr Urquhart (c) His mother (d) His classmates

7. What qualities helped Richard Ebright become a successful scientist?

Answers: 1. b 2. c 3. d 4. b 5. a 6. c 7. see notes